

Unit

6 Trigonometry

Sr. No.	Questions	A	B	C	D
1	The value of $\tan^{-1} 2$ in radians is:	$\frac{\pi}{2}$	$\frac{3\pi}{2}$	1.11π	$1.11\checkmark$
2	In a right triangle, the hypotenuse is 13 units and one of the angles is $\theta = 30^\circ$. The length of the opposite side is:	6.5 units $\checkmark$	7.5 units	6 units	5 units
3	A person standing 50 m away from a building sees the top of the building at an angle of elevation of 45° . Height of the building is:	50 m $\checkmark$	25 m	35 m	70 m
4	$\sec^2 \theta - \tan^2 \theta =$ _____.	$\sin^2 \theta$	$1\checkmark$	$\cos^2 \theta$	$\cot^2 \theta$
5	If $\sin \theta = \frac{3}{5}$ and θ is an acute angle, then $\cos^2 \theta =$	$\frac{7}{25}$	$\frac{24}{25}$	$\frac{16}{25}\checkmark$	$\frac{4}{25}$
6	$\left(\frac{5\pi}{24}\right) \text{ rad} =$ _____ degrees.	30°	$37.5^\circ\checkmark$	45°	52.5°
7	$292.5^\circ =$ _____ rad.	$\frac{17\pi}{6}$	$\frac{17\pi}{4}$	1.6π	$1.625\pi\checkmark$
8	Which of the following is a valid identity?	$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta\checkmark$	$\cos\left(\frac{\pi}{2} - \theta\right) = \cos \theta$	$\cos\left(\frac{\pi}{2} - \theta\right) = \sec \theta$	$\cos\left(\frac{\pi}{2} - \theta\right) = \csc \theta$
9	$\sin 60^\circ =$ _____.	1	$\frac{1}{2}$	$\sqrt{(3)^2}$	$\frac{\sqrt{3}}{2}\checkmark$
10	$\cos^2(100\pi) + \sin^2(100\pi) =$ _____.	$1\checkmark$	2	3	4

Solution of MCQs

1	Use calculator.
2	$\sin 30^\circ = \frac{\text{opposite or perp.}}{\text{hyp}}$ $\Rightarrow \text{opposite} = \text{hyp} \times \sin 30^\circ$ $= 13 \times 0.5$ $= 6.5$
3	$\tan 45^\circ = \frac{\text{perp}}{\text{base}}$ $\Rightarrow \text{perp} = \text{base} \times \tan 45^\circ$ $= 50 \times 1$ $\text{height} = 50$
4	$1 + \tan^2 \theta = \sec^2 \theta \Rightarrow \sec^2 \theta - \tan^2 \theta = 1$

5	$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ \cos^2 \theta &= 1 - \sin^2 \theta \\ &= 1 - \left(\frac{3}{5}\right)^2 \\ &= 1 - \frac{9}{25} \\ &= \frac{25 - 9}{25} \\ &= \frac{16}{25}\end{aligned}$
6	$\left(\frac{5\pi}{24}\right) rad = \left(\frac{5\pi}{24} \times \frac{180}{\pi}\right)^\circ = 37.5^\circ$
7	$292.5^\circ = \left(292.5 \times \frac{\pi}{180}\right) rad = \frac{13\pi}{8} = 1.625\pi$
8	$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$
9	Use calculator.
10	$\sin^2 \theta + \cos^2 \theta = 1$ Pythagorean identity for any θ

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